

Grade 10 Biology - Chapter 6: Photosynthesis (Lesson Notes)

Demonstration document - fictional data

Learning Objectives

By the end of this chapter students will be able to: state the balanced equation of photosynthesis; distinguish the light-dependent reactions from the Calvin cycle; explain the roles of chlorophyll, water, and carbon dioxide; and analyze how light intensity, temperature, and CO₂ concentration affect the rate of photosynthesis.

6.1 The Overall Equation

Photosynthesis converts light energy into chemical energy stored in glucose. The balanced equation is: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. The process occurs in chloroplasts, primarily in the palisade mesophyll cells of leaves.

6.2 Light-Dependent Reactions

These occur in the thylakoid membranes. Chlorophyll absorbs light, exciting electrons that pass along an electron transport chain. Water molecules are split (photolysis), releasing oxygen as a by-product. The energy captured is stored as ATP and NADPH, which power the next stage.

6.3 The Calvin Cycle

The Calvin cycle takes place in the stroma and does not require light directly. CO₂ is fixed by the enzyme RuBisCO onto a five-carbon acceptor (RuBP), producing three-carbon molecules that are reduced using ATP and NADPH into glyceraldehyde-3-phosphate (G3P). Some G3P leaves the cycle to form glucose; the rest regenerates RuBP.

6.4 Limiting Factors

At low light intensity, light is the limiting factor and the rate rises linearly with brightness. At high light intensity, CO₂ concentration or temperature becomes limiting. The rate roughly doubles for each 10 degree Celsius rise until about 40 degrees, beyond which enzymes denature and the rate collapses.

6.5 Common Misconceptions

Plants respire as well as photosynthesize - respiration continues day and night. Oxygen released comes from water, not from carbon dioxide. The Calvin cycle is not called the 'dark reaction' because it happens at night; it simply does not use light directly.